THE DERIVED UPPER SHRIEK FUNCTOR FOR A SEPARATED ETALE MORPHISM EQUALS INVERSE IMAGE

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Allowed standard prerequisites (editorial).

- Identification of the derived upper shriek with the locally quasi-finite shriek for separated quasi-finite morphisms of quasi-compact quasi-separated schemes, proved by uniqueness of adjoints using Rf_! = f_! (more-etale, Lemma lemma-Rf-upper-shriek-for-quasi-finite, canonical tag 0GL8). cited at proof lines 4098-4100. Named in the 'see for example' sentence as one of the two lemmas grounding the equality Rj_! = j_! in the adjoint-uniqueness argument.
- For an etale morphism the upper shriek equals inverse image, j^! = j^{-1} (more-etale, Lemma lemma-etale-upper-shriek, canonical tag 0F5A). cited at proof lines 4098-4100. Named in the same sentence; its proof gives that j_! is the left adjoint of j^{-1}, i.e. that j^{-1} is right adjoint to j_! for etale j.

Source reference. Original lemma 0GL8: sources/stacks-chow/tags.txt, source lines 20713–20713. Cited at proof lines 4098-4100: named lemma grounding the equality Rj_! = j_! in the adjoint-uniqueness argument.

Source reference. Original lemma 0F5A: sources/stacks-chow/tags.txt, source lines 18930–18930. Cited at proof lines 4098-4100: named lemma giving that j^{-1} is right adjoint to j_! for etale j.

Lemma 0.2. *Let $j : U \to X$ be a separated étale morphism of quasi-compact and quasi-separated schemes. Let Λ be a torsion coefficient ring. The functor $Rj^! : D(X_{étale}, \Lambda) \to D(U_{étale}, \Lambda)$ is equal to j^{-1}.*

Proof. This is true because both $Rj^!$ and j^{-1} are right adjoints to $Rj_! = j_!$. See for example Lemmas 0GL8 and 0F5A. □

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